

NeuronBench transfer from top-1, top-2, and “top5” evolution

Whole-frontier manual equation recovery on held-out tasks

Runs 313196, 313195, and 190177; prepared August 26, 2026

Result in one table

A held-out fit is counted as a *manual match* if at least one equation anywhere on its saved Pareto frontier expands, in physical units, to all required ground-truth monomials, contains no material extra monomial, and has clearly close coefficients. Tiny numerical artifacts are allowed. This is stricter about physical term support than merely calling a small NRMSE “solved,” but it can accept a structurally complete equation lying just above the fixed 10^{-6} NRMSE threshold.

Regime	evolution run	held-out tasks	matches	fits	manual rate
top-1	313196	5 worlds	19	25	76%
top-2	313195	4 worlds	16	20	80%
“top5” LOOCV	190177	z_rebound	5	5	100%

Training regime	held-out world	matches	fits	rate
top-1	h_sag	1	5	20%
	na_fatigue	5	5	100%
	ca_rebound	4	5	80%
	d_type	4	5	80%
	textbook_M	5	5	100%
top-2	na_fatigue	5	5	100%
	ca_rebound	4	5	80%
	d_type	4	5	80%
	textbook_M	3	5	60%
“top5”	z_rebound	5	5	100%

The top5 evaluation artifact is run 312780; it applies the final bundle from evolution run 190177. It evaluated all six worlds, but only **z_rebound** was absent during evolution, so only those five fits enter the LOOCV number above. All five best equations have the exact ground-truth polynomial support and no extra terms. Their best NRMSE values span 1.91×10^{-10} to 1.13×10^{-8} ; the largest absolute physical coefficient error is 1.17×10^{-4} (on a true coefficient of 480).

Interpretation. These are three individual evolution outcomes with different held-out sets and small denominators, not replicated estimates of a learning curve. In particular, “top5 LOOCV” currently means one completed fold (train on five, hold out **z_rebound**), not the aggregate of all six possible leave-one-world-out folds.

What NeuronBench asks PySR to recover

The benchmark converts six conductance-based neuron worlds into scalar symbolic-regression problems for the membrane vector field. Each row supplies the applied current I_{ext} , voltage V , and fully observed composite channel-open fractions ϕ_c ; the target is the exact current balance

$$\dot{V} = I_{\text{ext}} - \sum_c g_c \phi_c (V - E_c) - g_L (V - E_L).$$

Thus the difficult gating dynamics are exposed as inputs: the symbolic task is to reconstruct the additive ionic-current law, not to infer hidden gates from voltage alone.

For each world, the data contain 1,024 training and 16,384 test states. They are independent, reproducible scrambled-Sobol collocation states spanning $I_{\text{ext}} \in [-40, 20]$, $V \in [-95, 60]$, and each open fraction in $[0, 1]$; there is no observation noise. During a fit, $\hat{V}$ is divided by its training RMS and is unscaled before physical evaluation. PySR uses binary operators $\{+, -, \times\}$, no unary operators, maximum expression size 35, and up to 10^6 evaluations per final fit. The five final-evaluation seeds are 10000–10004.

Meta-evolution jointly changes four parts of PySR: mutation, population survival (which member is replaced), parent selection, and loss. Candidate bundle fitness is the fraction of training worlds for which a Pareto-frontier equation reaches $\text{NRMSE} \leq 10^{-6}$. Candidate bundles begin with three PySR seeds and the evolving population is reevaluated to ten seeds. The top-1 and top-2 runs used 15 generations with population 10 and 10 offspring per generation; the earlier top5 fold used 10 generations at the same population and offspring sizes.

Top-1 examples: solved, close, and not solved

The examples below all use the held-out **h_sag** world, so the comparison is like-for-like. Its physical ground truth is

$$f_*(I, V, \phi) = I_{\text{ext}} - 120\phi_{\text{Na}}(V - 50) - 36\phi_{\text{K}}(V + 77) - 0.3(V + 54.4) - 5\phi_{\text{Ih}}(V + 30).$$

Every column in Table 1 is a complete expanded equation $f = \sum_m c_m m$. A dash is a coefficient of zero, hence an absent monomial.

Table 1: Physical coefficients for representative top-1 Pareto equations.

Monomial m	ground truth	solved	close	not solved
I_{ext}	1	0.999964	1.00450	—
$V\phi_{\text{Na}}$	−120	−120.000008	−120.25974	−122.87369
ϕ_{Na}	6000	6000.00089	5986.04749	5488.86017
$V\phi_{\text{K}}$	−36	−35.999989	−36.14715	—
ϕ_{K}	−2772	−2771.99890	−2775.04212	−2442.69484
$V\phi_{\text{Ih}}$	−5	−4.999751	−5.14255	—
ϕ_{Ih}	−150	−149.96253	−152.41579	—
V	−0.3	−0.300133	—	−19.87721
1	−16.32	−16.34136	−6.59853	—

- **Solved (seed 10004, complexity 35, NRMSE 1.86×10^{-6}).** Every required current-balance monomial is present, there are no material extras, and the coefficients are extremely close. This is a manual match even though it narrowly misses the fixed numerical threshold.
- **Close but not solved (seed 10000, complexity 33, NRMSE 9.50×10^{-4}).** The Na, K, Ih, and applied-current terms are close, but the standalone leak-voltage term $-0.3V$ is absent. A shifted constant partially compensates on the sampled domain; algebraically it is not the full membrane equation.
- **Not solved (seed 10000, complexity 13 frontier point, NRMSE 9.09×10^{-2}).** This smaller expression has only part of the Na/K structure. It omits I_{ext} , the K voltage interaction, the entire Ih current, and the correct leak term. Its large standalone V coefficient cannot be interpreted as a small fitted perturbation.

Here “solved/close/not solved” describes manual structural inspection of the displayed frontier equations. It is intentionally distinct from the evaluation file’s exclusive numeric bins (recovered $\leq 10^{-6}$, near-exact $\leq 10^{-3}$, close ≤ 0.05 , miss otherwise).

Evolved operators

Top-1: run 313196

- **Mutation — compact_residual_linear_boost_mutation_gen13_0.** Evaluate the current tree, project its residual onto every raw feature, and add the one-dimensional least-squares correction αx_j with the largest predicted residual variance reduction. This is a direct additive-current “boosting” move.
- **Survival — age_regularized_survival.** Replace the oldest eligible member. This is the standard age-regularized baseline rather than a newly specialized neuron operator.

- **Selection** — `membrane_dynamics_selection_gen14_8`. Tournament cost is adjusted by complexity-frequency parsimony and a small penalty on older candidates, retaining newly discovered structures without allowing age to dominate fit.
- **Loss** — `adaptive_transition_weighted_mse_gen9_1`. A weighted MSE that emphasizes target extremity and large changes between adjacent rows, intended to highlight rare fast neuronal phases.

Top-2: run 313195

- **Mutation** — `mutate_residual_linear_step_gen12_6`. Randomly add either a residual-mean constant or a residual projection αx_j . Compared with top-1’s exhaustive feature choice, this makes smaller stochastic correction steps and can explicitly repair offsets.
- **Survival** — `age_regularized_survival`. Again, replace the oldest member.
- **Selection** — `frequency_robust_tournament_selection_gen4_2`. Tournament selection in log-cost space with a saturated frequency penalty for common complexities; this stabilizes the adaptive-parsimony calculation against extreme costs.
- **Loss** — `phase_aware_mse_tangent_loss_gen3_4`. Blends pointwise MSE with a normalized error in adjacent-row target/prediction differences, intended to favor local phase-space tangent agreement.

“Top5” LOOCV fold: run 190177

- **Mutation** — `residual_guided_additive_correction_gen3_2`. Select the feature most correlated with the residual and add its fitted OLS correction. It is an earlier, Pearson-correlation version of the residual-guided idea retained by top-1.
- **Survival** — `pareto_crowded_reverse_tournament_survival_gen5_4`. Prefer replacing candidates that are dominated, old, or crowded in complexity/log-loss space, while protecting isolated niche leaders.
- **Selection** — `pareto_rank_crowding_tournament_gen10_gen10_3`. Parent tournaments combine a non-domination rank, crowding bonus, and complexity-frequency novelty, encouraging coverage of multiple points along the loss-complexity frontier.
- **Loss** — `mse_loss`. Ordinary unweighted MSE; the five-world bundle kept the baseline loss and placed its specialization in search mechanics.

Across the three runs, the clearest repeated motif is *residual-guided additive mutation*, which matches the additive current-balance structure of the benchmark. Search diversity is handled differently: simple age turnover in top-1/top-2 versus explicit Pareto rank and crowding in top5. There is no equally consistent loss motif: top5 retained MSE while the smaller-training runs evolved dynamics-themed weighting.

One important caveat is that Sobol collocation rows are not consecutive samples from a neuron trajectory. Therefore “adjacent-row transitions” in the top-1 loss and “tangents” in the top-2 loss do not represent physical time derivatives under this dataset ordering. Target extremity weighting remains meaningful, but any advantage attributed specifically to temporal phase consistency would require a trajectory-ordered benchmark or an order-invariance test.

Artifacts and audit trail

The top-1/top-2 whole-frontier decisions come from `reports/neuron_manual_match_comparison.json`; the source frontiers are `runs/313196/neuron_full_eval/neuron_results.json` and `runs/313195/neuron_full_eval/neuron_results.json`. The top5 frontiers are in `runs/312780/neuron_results.json`, linked to bundle source `runs/190177`; their physical expansions and support checks are recorded in `reports/neuron_top5_symbolic_recovery_analysis.json`. Operator implementations and run configurations are stored in each evolution run’s `run_data.json`.